

Shell & Bash Commands for 1337 / 42 (Shell00 + Shell01)

1. Navigation

Command	Purpose
<code>pwd</code>	Show current directory
<code>ls</code>	List files and directories
<code>ls -l</code>	Long listing format
<code>ls -la</code>	Show all files (including hidden)
<code>cd dir</code>	Enter a directory
<code>cd ..</code>	Go to parent directory
<code>cd ~</code>	Go to home directory

2. File & Directory Management

Command	Purpose
<code>touch file</code>	Create an empty file
<code>mkdir dir</code>	Create a directory
<code>cp src dst</code>	Copy file or directory
<code>mv src dst</code>	Move or rename
<code>rm file</code>	Remove file
<code>rm -r dir</code>	Remove directory recursively

3. File Content

Command	Purpose
<code>cat file</code>	Display file content

Command	Purpose
<code>head file</code>	Show first lines
<code>tail file</code>	Show last lines
<code>wc file</code>	Count lines, words, bytes

4. Search & Filtering

Command	Purpose
<code>grep pattern file</code>	Search text
<code>find path</code>	Search files/directories
<code>sort</code>	Sort lines
<code>uniq</code>	Remove duplicate consecutive lines

5. Permissions

Permission Values

Permission	Value
<code>r</code> (read)	4
<code>w</code> (write)	2
<code>x</code> (execute)	1

Common Modes

Mode	Meaning
<code>777</code>	<code>rwX rwX rwX</code>
<code>755</code>	<code>rwX r-X r-X</code>
<code>700</code>	<code>rwX --- ---</code>

Mode	Meaning
644	rW- r-- r--
600	rW- --- ---

Commands

Command	Purpose
<code>chmod mode file</code>	Change permissions
<code>chown user file</code>	Change owner

Examples:

❏ Bash

```
chmod 755 file
chmod 644 file
chmod 600 file
```



6. Links

Command	Purpose
<code>ln file link</code>	Create hard link
<code>ln -s file link</code>	Create symbolic link

7. Archives

Command	Purpose
<code>tar</code>	Create/extract archives
<code>gzip</code>	Compress files
<code>gunzip</code>	Extract <code>.gz</code> files

8. Command Information & Help

Command	Purpose
<code>man cmd</code>	Full manual
<code>help cmd</code>	Help for builtin commands
<code>type cmd</code>	Show command type
<code>which cmd</code>	Show executable path
<code>whatis cmd</code>	Short description

Quick Difference

Command	Use
<code>man</code>	Detailed documentation
<code>help</code>	Builtin command help
<code>type</code>	Is it builtin, alias, function, or executable?
<code>which</code>	Where is the executable located?
<code>whatis</code>	One-line description

9. Redirections & Pipes

Symbol	Purpose
<code>></code>	Redirect output (overwrite)
<code>>></code>	Redirect output (append)
<code><</code>	Redirect input

Examples:

```
</> Bash
```



```
ls -l > files.txt
```

❏ Bash



```
cat file.txt | grep hello
```

10. Special Paths

Symbol	Meaning
.	Current directory
..	Parent directory
~	Home directory
/	Root directory

11. Environment Variables

Command	Purpose
env	Show environment variables
echo \$VAR	Show variable value
export VAR=value	Create/export variable

Examples:

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```
echo $HOME
```

❏ Bash



```
export NAME=yassine
```

12. Essential Commands for Shell00 + Shell01

⌕ Bash



```
pwd
ls
cd
mkdir
touch
cp
mv
rm
cat
grep
find
chmod
chown
ln
tar
gzip
gunzip
man
help
type
which
whatis
echo
env
export
sort
uniq
wc
```

Essential Symbols

⌕ Bash



```
.
..
~
/
>
>>
<
|
*
```